

PROGRAM 7: DEPTH FIRST SEARCH (DFS)

Aim

To write a Python program to implement Depth First Search (DFS) traversal of a graph.

Objective

To traverse all the vertices of a graph using the Depth First Search (DFS) algorithm.

Algorithm

1. Start.
2. Create a graph using an adjacency list.
3. Mark all vertices as unvisited.
4. Visit the starting vertex and print it.
5. Mark the vertex as visited.
6. Recursively visit all unvisited adjacent vertices.
7. Repeat until all vertices are visited.
8. Stop.

Source Code

```
graph = {  
    'A': ['B', 'C'],  
    'B': ['D', 'E'],  
    'C': ['F'],  
    'D': [],  
    'E': [],  
    'F': []
```

```
}
```

```
visited = []
```

```
def dfs(node):
```

```
    if node not in visited:
```

```
        print(node, end=" ")
```

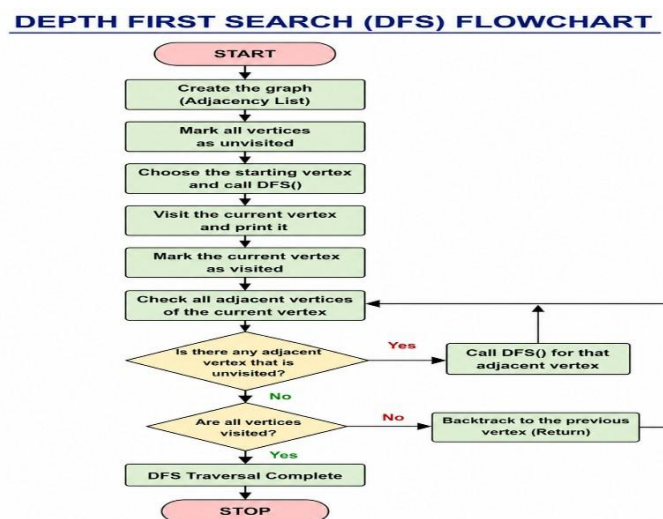
```
        visited.append(node)
```

```
    for i in graph[node]:
```

```
        dfs(i)
```

```
dfs('A')
```

Flowchart



Sample Output

A B D E C F

```
>>> |  
    |==== RESTART: C:/Use  
    |A B D E C F  
>>> |
```

Result

Thus, the Python program to implement Depth First Search (DFS) was executed successfully.